

CYBERSECURITY PORTFOLIO

Cybersecurity Lab Report

Kernel command line and boot messages

Environment: Kali Linux (virtual machine)

Labs included: 1

Date: October 2026

CONTENTS

Lab Overview

This report documents a hands-on lab performed on Kali Linux: chapter, topic and objective, followed by a step-by-step walkthrough with screenshots.

LAB 3

Kernel Command Line and Boot Messages

Chapter	Lab 5 (chapter name: add here)
Topic	Kernel boot parameters and boot messages
Objective	Read the kernel command line and filter boot messages with dmesg.

LAB 3

Kernel Command Line and Boot Messages

Chapter: Lab 5 | Topic: Kernel boot parameters and dmesg

Objective

Read the parameters the kernel was booted with and look at the early boot messages. Environment: Kali Linux VM (VirtualBox), terminal.

Procedure

1 Read the kernel command line

```
(kali@kali)-[~]
$ cat /proc/cmdline
BOOT_IMAGE=/boot/vmlinuz-6.19.14+kali-amd64 root=UUID=aaaa6263-4f91-4dad-b5b1-825f7867e619 ro quiet splash

(kali@kali)-[~]
$ cat /proc/cmdline | tr ' ' '\n'
BOOT_IMAGE=/boot/vmlinuz-6.19.14+kali-amd64
root=UUID=aaaa6263-4f91-4dad-b5b1-825f7867e619
ro
quiet
splash

(kali@kali)-[~]
$ cat /proc/cmdline | grep -E 'quiet|splash|root|ro|rw'
BOOT_IMAGE=/boot/vmlinuz-6.19.14+kali-amd64 root=UUID=aaaa6263-4f91-4dad-b5b1-825f7867e619 ro quiet splash

(kali@kali)-[~]
$ dmesg | head -30
[ 0.000000] Linux version 6.19.14+kali-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-15 (Debian 15.2.0-17) 15.2.0
, GNU ld (GNU Binutils for Debian) 2.46) #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05)
[ 0.000000] Command line: BOOT_IMAGE=/boot/vmlinuz-6.19.14+kali-amd64 root=UUID=aaaa6263-4f91-4dad-b5b1-825f7867
e619 ro quiet splash
[ 0.000000] BIOS-provided physical RAM map:
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
[ 0.000000] BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000000a0000-0x00000000000fffff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000007fffff] usable
[ 0.000000] BIOS-e820: [mem 0x00000000007fff0000-0x00000000007fffff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x00000000fec00000-0x00000000fec0ffff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000fee00000-0x00000000fee0ffff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000ffff0000-0x00000000ffffffffff] reserved
[ 0.000000] NX (Execute Disable) protection: active
[ 0.000000] APIC: Static calls initialized
[ 0.000000] SMBIOS 2.5 present.
[ 0.000000] DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
[ 0.000000] DMI: Memory slots populated: 0/0
[ 0.000000] Hypervisor detected: KVM
[ 0.000000] last_pfn = 0x80000 max_arch_pfn = 0x400000000
[ 0.000000] kvm-clock: Using msrs 4b564d01 and 4b564d00
[ 0.000003] kvm-clock: using sched offset of 9773636522 cycles
[ 0.000005] clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle_ns: 88159059148
3 ns
[ 0.000009] tsc: Detected 2803.198 MHz processor
[ 0.001594] e820: update [mem 0x00000000-0x00000fff] usable => reserved
Activate Windows
Go to Settings to activate Windows.
```

Boot parameters, one per line, highlighted, then the first kernel messages.

/proc/cmdline holds the boot parameters. tr puts each on its own line, grep -E highlights selected ones and dmesg | head -30 shows the first kernel messages.

Parameter	Meaning
BOOT_IMAGE=/boot/vmlinuz-6.19.14+kali-amd64	Kernel image that was loaded
root=UUID=...	Device used as the root filesystem
ro	Root filesystem is mounted read-only at first
quiet	Reduces boot messages on screen
splash	Shows the boot splash screen

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Filter the boot messages

```

$ dmesg | grep -iE 'boot|kernel|command'
0.000000] Command line: BOOT_IMAGE=/boot/vmlinuz-6.19.14+kali-amd64 root=UUID=aaaa6263-4f91-4dad-b5b1-825f7867
0.021186] Booting paravirtualized kernel on KVM
0.026688] Kernel command line: BOOT_IMAGE=/boot/vmlinuz-6.19.14+kali-amd64 root=UUID=aaaa6263-4f91-4dad-b5b1-8
0.102799] mitigations: Enabled attack vectors: user_kernel, user_user, guest_host, guest_guest, SMT mitigation
0.116316] Yama: disabled by default; enable with sysctl kernel.yama.*
0.224656] smpboot: CPU0: 11th Gen Intel(R) Core(TM) i7-1165G7 @ 2.80GHz (family: 0x6, model: 0x8c, stepping: 0
0.226630] smpboot: Total of 1 processors activated (5606.39 BogomIPS)
0.228316] Memory: 1940300K/2096696K available (17882K kernel code, 3350K rwdata, 13036K rodata, 4476K init, 55
0.228754] DMA: preallocated 256 KiB GFP_KERNEL|GFP_DMA pool for atomic allocations
0.228773] DMA: preallocated 256 KiB GFP_KERNEL|GFP_DMA32 pool for atomic allocations
0.301380] pci 0000:00:02.0: vgaarb: setting as boot VGA device
0.886830] Loaded X.509 cert 'Build time autogenerated kernel key: 42bf05ba17f96a2ee677e0d164c07695b0bbcb36'
0.982095] Freeing unused kernel image (initmem) memory: 4476K
0.982119] Write protecting the kernel read-only data: 32768k
0.982294] Freeing unused kernel image (text/rodata gap) memory: 548K
0.982492] Freeing unused kernel image (rodata/data gap) memory: 1300K
1.788764] ahci 0000:00:0d.0: AHCI vers 0001.0100, 32 command slots, 3 Gbps, SATA mode
2.411278] vmwgfx 0000:00:02.0: [drm] Capabilities: rect copy, cursor, cursor bypass, cursor bypass 2, alpha cu
cursor, extended fifo, pitchlock, irq mask, gmr, traces, gmr2, screen object 2, command buffers, command buffers 2, g
object, cap2 register,
2.534548] vmwgfx 0000:00:02.0: [drm] Using command buffers with DMA pool.
4.758355] systemd[1]: Listening on systemd-udev-kernel.socket - udev Kernel Socket.
4.770774] systemd[1]: Mounting sys-kernel-debug.mount - Kernel Debug File System...
4.778127] systemd[1]: Mounting sys-kernel-tracing.mount - Kernel Trace File System...
4.778193] systemd[1]: auth-rpcgss-module.service - Kernel Module supporting RPCSEC_GSS skipped, unmet conditio
check ConditionPathExists=/etc/krb5.keytab
4.799698] systemd[1]: Starting modprobe@configfs.service - Load Kernel Module configfs...
4.799742] systemd[1]: modprobe@drm.service - Load Kernel Module drm skipped, unmet condition check ConditionKe
KernelModuleLoaded=drm
4.812599] systemd[1]: Starting modprobe@efi_pstore.service - Load Kernel Module efi_pstore...
4.812636] systemd[1]: modprobe@fuse.service - Load Kernel Module fuse skipped, unmet condition check Condition
KernelModuleLoaded=fuse
4.876603] systemd[1]: Starting systemd-modules-load.service - Load Kernel Modules...
4.885747] systemd[1]: Starting systemd-remount-fs.service - Remount Root and Kernel File Systems...
4.905686] systemd[1]: Mounted sys-kernel-debug.mount - Kernel Debug File System.
4.905739] systemd[1]: Mounted sys-kernel-tracing.mount - Kernel Trace File System.
4.960177] systemd[1]: Finished modprobe@configfs.service - Load Kernel Module configfs.
4.960399] systemd[1]: Finished modprobe@efi_pstore.service - Load Kernel Module efi_pstore.
4.960550] systemd[1]: Finished systemd-modules-load.service - Load Kernel Modules.
4.960692] systemd[1]: Finished systemd-remount-fs.service - Remount Root and Kernel File Systems.
4.963693] systemd[1]: Mounting sys-kernel-config.mount - Kernel Configuration File System.
5.024602] systemd[1]: Starting systemd-sysctl.service - Apply Kernel Variables...

```

dmesg filtered for boot, kernel and command.

`dmesg | grep -iE 'boot|kernel|command'` keeps only matching lines. The command line appears again at the start of the log, followed by systemd setting up kernel modules and file systems.

Results and Takeaways

- The kernel was booted from `vmlinuz-6.19.14+kali-amd64` with `ro quiet splash`.
- `dmesg` shows hardware detection and boot steps, including the VirtualBox environment.
- `/proc/cmdline` and `dmesg` are the first places to check how a system booted.